

AI FUNDAMENTALS SERIES · PART 1 OF 3 · SECTIONS 01 TO 05

AI Understanding Basics

The Core Taxonomy, Learning Mechanics, Parameters, GPU Compute & Model Evolution

Instructor & Student Guide · 5 Foundational Chapters · Engineering Foundations

Part 1 Overview: This document covers the foundational core of artificial intelligence: how the disciplines fit together, how mathematical models learn by minimizing loss, what billions of parameters physically mean in GPU memory, the 7-stage manufacturing production line, and how models are improved through pre-training vs fine-tuning.

1 01. The Map of AI: Hierarchy & Core Paradigms

AI is not one single technology. It is a strict hierarchy of ideas. Most confusion in the industry comes from using one word (“AI”) for four completely different conceptual layers.

1. ARTIFICIAL INTELLIGENCE (AI)

The umbrella term: Any machine performing tasks that simulate human cognitive abilities (Search Algorithms, Logic Engines, Expert Rules, Robotics, NLP, and Machine Learning).

2. MACHINE LEARNING (ML)

A subset of AI: Systems that learn statistical patterns from data automatically, rather than being explicitly programmed with human hand-written rules.

3. DEEP LEARNING (DL)

A subset of ML: Multi-layered Artificial Neural Networks (> 3 hidden layers). Discovers hierarchical representations directly from raw unstructured data.

4. GENERATIVE AI & LLMs

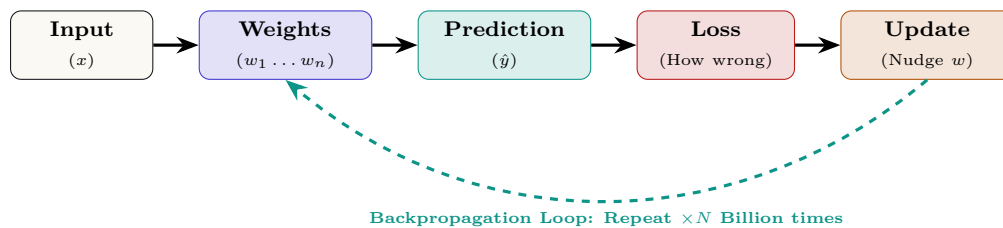
Specialized frontier DL models trained to generate new synthetic text, code, images, and audio (e.g., GPT-4, Claude, Llama).

1.1 Rule-Based Systems vs Learning-Based Systems

Dimension	Rule-Based Systems (GOF AI / Classical)	Learning-Based Systems (ML / DL)
Logic Creator	A human programmer writes every single rule.	The machine derives statistical rules from examples.
Input Flow	Data + Rules $\implies$ Answers	Data + Trained Model (Rules) $\implies$ Answers
Code Example	<code>if temperature > 30: turn_on(ac)</code>	50,000 photos $\rightarrow$ learn representation $\rightarrow$ predict
Fragility	Fails on edge cases not anticipated in code.	Handles messy, high-variety unstructured inputs.
Auditability	100% predictable and line-by-line auditable.	Logic ends up as millions of unreadable floating-point weights.

2 02. How a Model Learns: The Training Loop & Loss

A model does not “understand” the world the way humans do. It adjusts numerical weights until its predictions get mathematically less wrong.



2.1 Memorization ≠ Learning: Database vs Model

Dimension	Traditional Database (Postgres / SQL)	Trained Neural Model (Weights)
What It Stores	Exact raw rows, records, and text.	Fixed floating-point numbers representing statistical patterns.
Retrieval	Deterministic: returns exact matches only.	Generative: synthesizes plausible responses.
Generalization	Cannot answer anything outside its rows.	Can answer beyond training rows, but can be factually wrong.

GARBAGE IN, GARBAGE PARAMETERS

Bad Data → Bad Statistical Patterns → Bad Model → Confident Nonsense in Production

Engineering Reality: In production, when a fine-tune or RAG pipeline underperforms, the fix is almost always in the **dataset quality and cleaning**, not the model architecture!

3 03. Billions of Parameters: Hardware Economics & Tradeoffs

***What is a Parameter?** A parameter is a learned numerical value (weight/bias) inside a neural network. It is not a fact or a sentence. In modern models, each parameter is stored as a 16-bit float (FP16 $\approx$ 2 bytes).*

3.1 Model Scale & Hardware Memory Math

Size Class	Exact Parameter Count	VRAM (FP16 / 16-bit)	Deployment Hardware & Best Use Case
Small (7B)	$\approx 7,000,000,000$	≈ 14 GB	Runs on a single consumer GPU (e.g., RTX 4090 / Mac M-series). Ideal for fast classification, extraction, routing.
Mid-Size (70B)	$\approx 70,000,000,000$	≈ 140 GB	Requires a dedicated multi-GPU server (e.g., 4 $\times$ or 8 $\times$ A100/H100). Balanced everyday enterprise work.
Frontier (405B+)	$\approx 405,000,000,000$	≈ 810 GB	Requires sharded high-speed GPU clusters. Broadest reasoning, complex coding, open-ended problem solving.

3.2 The 3 Tradeoff Dimensions of Scaling Up

- **Capability:** Climbs rapidly with scale, but eventually encounters diminishing returns (plateaus).
- **Inference Cost:** Increases strictly linearly ($1\times \rightarrow 10\times \rightarrow 60\times$) with parameter size.
- **Latency:** Larger models require more memory bandwidth and parallel GPU sync, increasing time-to-first-token.

[!] MORE PARAMETERS $\neq$ AUTOMATICALLY BETTER FOR EVERY TASK

Frontier models cost orders of magnitude more per token and have higher latency. For a narrow, well-defined task (e.g., sorting support tickets), a fine-tuned 7B model will often be **faster, cheaper, and more accurate** than an untuned 405B frontier model.

4 04. From Data to Model: The 7-Stage Manufacturing Pipeline

A large foundation model is not written in code. It is manufactured on a 7-stage industrial computing production line.



4.1 Detailed Breakdown of the 7 Stations

1. **Data Collection (Petabytes):** Web crawls, books, licensed corpora, code, synthetic data.
2. **Data Cleaning & Filtering:** Discarding > 80% of raw crawls (deduplication, safety, boilerplate).
3. **Tokenization (Text → Integer IDs):** Subword vocabulary mapping words to tokens.
4. **Pre-training (Next-Token Objective):** Massive multi-million dollar compute run across thousands of GPUs.
5. **Evaluation & Benchmarks:** Systematic safety testing, MMLU benchmarks, and human preference scoring.
6. **Fine-Tuning & Post-Training:** Instruction tuning, RLHF, and safety alignment.
7. **Deployment & Inference:** Quantization (FP16 → INT8/INT4), KV caching, batching, and serving APIs.

4.2 Why GPUs? Checkpoints & The Knowledge Cutoff

- **Why GPUs:** Training is a single mathematical operation repeated trillions of times: **matrix multiplication and addition**. A CPU has a few powerful cores; a GPU has thousands of simple arithmetic units executing parallel matrix ops simultaneously.
- **Checkpoints:** Snapshots of weights saved during training. If cluster nodes crash mid-run, training resumes from the last healthy checkpoint.
- **Knowledge Cutoff:** Training data ceases at a specific date. The model has zero awareness of events after this cutoff. **To provide real-time facts, use Search, Tools, or RAG at request time.**

5 05. Models Don't Just Get Smarter: Retraining vs Fine-Tuning

A model does not automatically know what happened yesterday. When you want better performance, you have distinct architectural paths.

Approach	Training From Scratch (Pre-training)	Fine-Tuning (LoRA / SFT)
Starting State	Random initialization ($w = \text{random}$).	Pre-existing base model with pre-learned weights.
Data Needed	Trillions of tokens (Terabytes/Petabytes).	Thousands of curated pairs (Megabytes).
Compute Cost	Millions of dollars; weeks on GPU clusters.	Tens to hundreds of dollars; hours on 1–4 GPUs.
Primary Goal	Create a new general foundation model.	Teach style, tone, output format, and task domain .

[!] THE COMMON INDUSTRY MISCONCEPTION ABOUT FINE-TUNING

Fine-tuning is NOT an effective way to teach a model new facts!

- Fine-tuning is brilliant for **form and behavior** (e.g., “Always respond in valid JSON with these 4 keys” or “Adopt a medical doctor persona”).
- For teaching **private factual knowledge** (e.g., your company’s internal PDFs or documentation), **RAG (Retrieval-Augmented Generation)** is **faster, cheaper, verifiable, and does not hallucinate outdated weights**.

5.1 The 6 Ways a Production AI System Actually Improves

Level	Improvement Mechanism	What Actually Happens
01	New Pre-training	A complete fresh pre-training run on larger, higher-quality data.
02	Domain Fine-Tuning	Targeted parameter tuning for specific syntax or formats.
03	Post-Training / RLHF	Steering model preferences via human or AI feedback loops.
04	Higher Quality Data	Cleaning the existing corpus (better data beats bigger models).
05	Evaluation Suites	Designing strict automated tests to pinpoint failure modes.
06	System Design (RAG & Tools)	Orchestration: adding vector search, caching, tools, and guardrails. (90% of engineering wins happen here!).

Module 1 Master Summary Spine

QUICK REVISION & MASTER TAKEAWAYS

